

Study Questions: Systems

1. What is a cortical column, and how can it be understood as a local inference engine within the Active Inference framework?
2. Describe the hierarchical organization of the neocortex. How does this hierarchy map onto the depth of the generative model?
3. What constitutes the Markov Blanket of the brain? Why is it the sensory-motor surface of the body rather than the skull?
4. Explain the brain-body-environment loop as the physical implementation of Active Inference's perception-action cycle.
5. What did Friston (2005) demonstrate about resting-state brain activity? How does this relate to the concept of a generative model?
6. Name the three major resting-state networks discussed in this module. What role might each play in the brain's generative model?
7. How might the default mode network (DMN) encode the agent's self-model? What evidence supports this interpretation?
8. What is the proposed role of the salience network in precision weighting? How does the anterior insula contribute?
9. How does the concept of prediction error manifest at the neural level? What are the candidate neural signals?
10. Compare "bottom-up" prediction errors with "top-down" predictions in the cortical hierarchy. Which cortical layers are associated with each?
11. What is anosognosia, and how does Active Inference explain it as a failure of generative model updating?
12. How might depersonalization be understood as low precision on self-model predictions?
13. What experimental paradigm could test whether resting-state activity reflects an active generative model rather than mere neural noise?
14. How do sensory cortices (V1, A1, S1) differ from association cortex in terms of the generative model hierarchy?
15. What role do thalamocortical loops play in the brain's implementation of hierarchical inference?

16. How does the concept of neural Darwinism (Edelman) relate to the self-organizing properties of cortical assemblies?
17. What is the relationship between neural oscillation frequencies (alpha, beta, gamma) and the timescale hierarchy of the generative model?
18. How does the brain's metabolic budget (consuming ~20% of body energy at ~2% body mass) reflect the computational demands of maintaining a generative model?
19. Compare the Active Inference account of resting-state activity with the "default mode" interpretation (Raichle, 2001). Are they compatible?
20. Design a hypothetical fMRI study that would test a specific prediction of Active Inference about neural self-organization.